

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	18 june 2026
Name	Mohit
Project Name	Shop EZ
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution

FR	Functional Requirement	Sub Requirement (Story / Sub-Task)
No.	(Epic)	
FR-1	User Registration	Registration through FormRegistration through Gmail
FR-2	User Confirmation	Confirmation via EmailAccount Activation
FR-3	User Login & Authentication	Login through Email & PasswordForgot PasswordPassword Reset
FR-4	User Profile Management	View ProfileUpdate Profile InformationChange Password
FR-5	Product Catalog Management	View ProductsBrowse CategoriesView Product Details
FR-6	Product Search & Filtering	Search by Product NameFilter by CategorySort by FR-Price
FR-7	Shopping Cart Management	Add Product to CartUpdate QuantityRemove Product from Cart

FR-8 Wishlist Management	Add Products to Wishlist Remove Products from Wishlist
FR-9 Order Management	Place Order View Order History Track Order Status
FR No. Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR10 Payment Processing	Online Payment via Razorpay Payment Confirmation
FR11 Email Notifications	Registration Confirmation Email Order Confirmation Email Order Status Updates
FR12 Product Management (Admin)	Add Product Edit Product Delete Product
FR13 Inventory Management (Admin)	Update Stock Quantity Monitor Inventory Levels
FR14 Customer Management (Admin)	View Registered Users Manage User Accounts
FR15 Sales & Reports (Admin)	View Sales Reports Generate Order Reports
FR16 Customer Support	Submit Support Request View Support Responses
FR17 Security & Access Control	Secure Authentication Role-Based Access Control (Admin/User) Password Encryption
Product	Recommend Popular Products Display Related
FR18 Recommendation System Products	

NON FUNCTIONAL REQUIREMENTS

NFR	Non-Functional	Description	No.
	Requirement		

NFR-1	Usability	The application shall provide an intuitive, user-friendly, and responsive interface that enables users to browse products, place orders, and manage accounts with minimal training.
NFR-2	Security	The system shall protect user data through secure authentication, password encryption (bcrypt), HTTPS communication, and role-based access control for administrators and customers.
NFR-3	Reliability	The application shall consistently process user requests, orders, and payments with minimal errors and maintain data integrity even during unexpected failures.
NFR-4	Performance	The system shall load pages quickly, process user requests efficiently, and support multiple concurrent users without noticeable delays through optimized APIs and database indexing.
NFR-5	Availability	The application shall be accessible 24/7 through cloud deployment and maintain high uptime to ensure uninterrupted access for customers and administrators.
NFR-6	Scalability	The architecture shall support future growth by allowing additional users, products, and transactions without significant changes to the system design.
NFR-7	Maintainability	The system shall be developed using modular and well-documented code to simplify updates, debugging, and feature enhancements.
NFR-8	Compatibility	The application shall function correctly across major web browsers such as Chrome, Edge, Firefox, and Safari, and support desktop and mobile devices.
NFR-9	Data Integrity	The system shall ensure accurate storage, retrieval, and updating of user, product, and order information in the database.

NFR- 10	Recoverability	The application shall support database backup and recovery mechanisms to prevent data loss in case of system failures.
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